MARK SCHEME · TEACHER COPY

11BR · Topic 3: Hardware

Cambridge AS Level Computer Science 9618

Paper 1 Theory Fundamentals

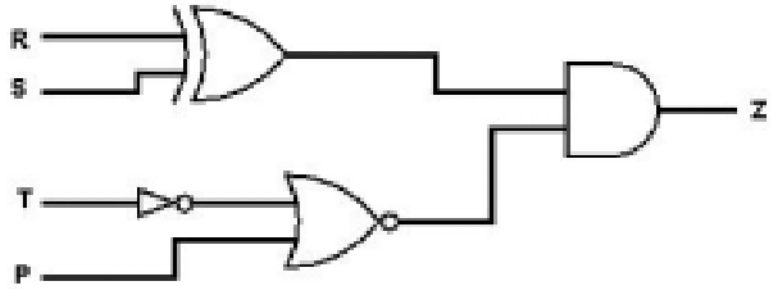
Total marks: 75

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

For teacher / admin use only. Do not give to students before they have attempted the paper.

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Question 1 · May/June 2023 Paper 12, original Q6 (a),(b) [4 marks]

Question	Answer	Marks
1(a)	**1 mark** for correct XOR and AND gates, **with** correct inputs **1 mark** for correct NOT and NOR gates **with** correct inputs  	2
1(b)	**1 mark** for first 4 rows **1 mark** for last 4 rows	2

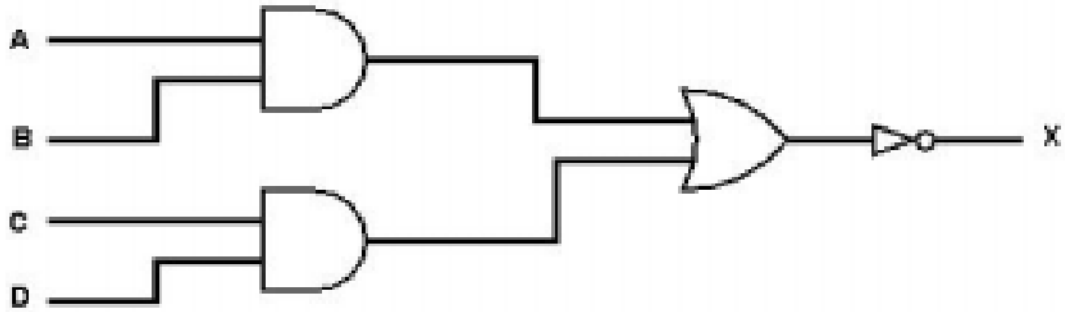
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1	0	1		0

Total: 75 marks

				1	1	0				1				
				1	1	1				1				

Question 2 · May/June 2024 Paper 12, original Q1 (a),(b) [6 marks]

Question	Answer	Marks
2(a)	**1 mark** for each correct answer: *NAND* • The output is 0 when both inputs are 1, otherwise the output is 1 *NOR* • The output is 1 when both inputs are 0, otherwise the output is 0 *XOR* • The output is 1 when one of the inputs is 1 and the other input is 0, otherwise the output is 0 *OR* • The output is 0 when both inputs are 0, otherwise the output is 1	**4**

2(b)	**1 mark** for both AND gates with correct inputs **1 mark** for correct OR and NOT gates with correct inputs and no superfluous gates: 	**2**

Question 3 · May/June 2025 Paper 12, original Q4 (a),(b) [4 marks]

Question	Answer	Marks
3(a)	**1 mark** each: • X = (R NOR S) XOR T • Y = (T AND U) OR U	**2**
3(b)	**1 mark** for first four rows, **1 mark** for second four rows	**2**

A	B	C	Working space	X
0	0	0		1
0	0	1		1
0	1	0		1

Total: 75 marks

0	1	1		**1**
1	0	0		**1**
1	0	1		**0**
1	1	0		**0**
1	1	1		**1**

Question 4 · May/June 2025 Paper 13, original Q4 (a),(b) [4 marks]

Question	Answer	Marks
4(a)	**1 mark** for each correct statement X = B NOR (A NOR C) Y = (B AND C) XOR NOT D	**2**
4(b)	**1 mark** for 1st 4 rows **1 mark** for 2nd 4 rows	**2**

A	B	C	Working space	X
0	0	0		**1**
0	0	1		**1**
0	1	0		**1**
0	1	1		**0**

1	0	0		**0**
1	0	1		**1**
1	1	0		**1**
1	1	1		**1**

Question 5 · May/June 2026 Paper 13, original Q8 (a),(b),(c) [10 marks]

Question	Answer	Marks	Guidance
5(a)	**1 mark** for name, **1 mark** for a matching description • EPROM • Erase using UV rays // needs removing to erase // have to erase whole device // supports many read/write cycles • EEPROM • Erase using electric signals // erase in situ // can update parts of memory // uses floating gate technology // supports many read/write cycles	**4**	No mark for expanding acronym for description.
5(b)	**1 mark** each. • The microphone has a diaphragm / ribbon • The incoming sound **waves** cause **vibrations** of the diaphragm • … causing a coil to move past a magnet // causing a magnet to move past a coil (dynamic microphone). • // changing the capacitance (condenser microphone) // deforms the crystal (crystal microphone) • An electrical signal is produced	**3**	For MP1 accept diaphragm equivalent e.g. cone.
5(c)	**1 mark** each • Temporary storage of data • …until the device is ready to receive it // until a sound player takes the data • Data may be sent faster than it can be received // Allows for different speed of transmission between devices • Allows processor to carry out other tasks	**3**	

Question 6 · Oct/Nov 2023 Paper 12, original Q1 (a),(b),(c) [6 marks]

Question	Answer	Marks
6(a)	**1 mark** for: • infra-red / proximity (sensor)	1
6(b)	**1 mark** for each bullet point (**max 2**) • Actuator generates a signal / causes an action / converts electrical energy into a mechanical force • ... to push an arm // to open a trap door // to pick up the chocolate bar with the incorrect weight	2
6(c)(i)	**1 mark** for each feature (**max 2**) • Dedicated to a single task // limited number of functions • Built into a larger system // integrated into a larger system • An embedded system must contain a processor, memory and an I/O capability // Dedicated hardware	2
6(c)(ii)	**1 mark** for each bullet point (**max 1**) • Difficult to change / update the firmware by the user // Difficult to upgrade devices to take advantage of new technology • Cannot be easily **adapted** for **another** task • Troubleshooting faults/upgrading is a specialist task • Devices are often thrown away rather than repaired as difficult to upgrade or repair	1

Question 7 · Oct/Nov 2023 Paper 12, original Q4 (a),(b) [5 marks]

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>7(a)</td><td>1 mark for each bullet point

• A NAND B
• NOT(B XOR C)
• Final NAND

X = A NAND B NAND (NOT(B XOR C))</td><td>3</td></tr>
<tr><td>7(b)</td><td>1 mark for each set of rows (shaded)

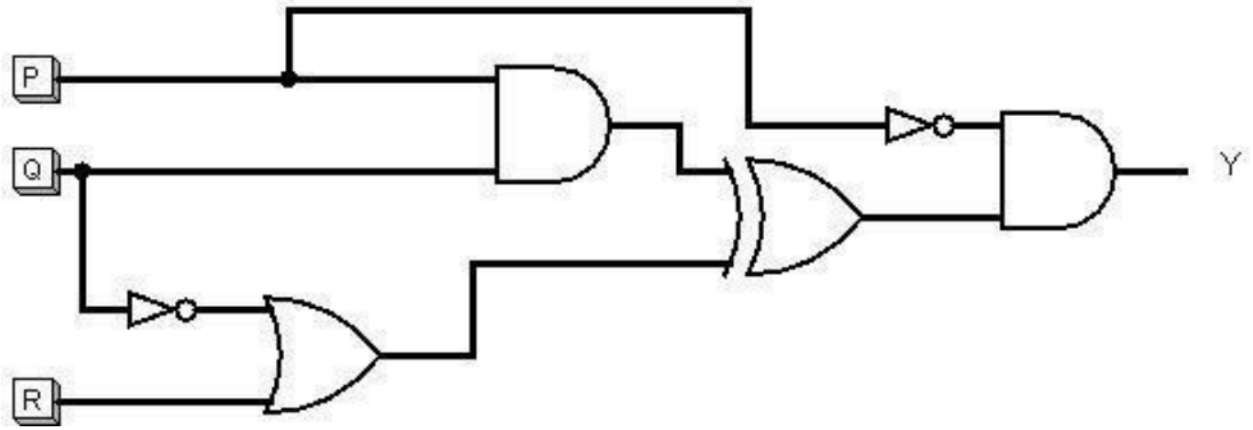
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</td><td>2</td></tr>
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Total: 75 marks

Question 8 · Oct/Nov 2023 Paper 13, original Q4 (a),(b) [4 marks]

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>8(a)</td><td>1 mark for each set of highlighted rows

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<tr><td>1</td><td>1</td><td>1</td><td>0</td></tr>
</table></td><td>2</td></tr>
<tr><td>8(b)</td><td>1 mark for P AND Q <u>and</u> NOT Q OR R
1 mark for NOT gate and AND gate and XOR gate

 
</td><td>2</td></tr>
</table>

Question 9 · Oct/Nov 2023 Paper 13, original Q7 (a),(b),(c),(d) [11 marks]

Question	Answer	Marks
9(a)	**1 mark** for each bullet point (**max 2**). • Static RAM has faster access time • …because it does not need to be refreshed • …used on the CPU for improvement of CPU cache speed • Static RAM has lower	**2**
9(b)	**1 mark** for each bullet point (**max 2**) • EEPROM allows frequent / multiple read / write / erase operations • …which means that the contents of the firmware in the headset can be changed easily • …without fully erasing the contents of the firmware in the headset first	**2**
9(c)	**1 mark** for each bullet point (**max 4**) • The print instructions and data are sent by the laptop to a buffer (at laptop speed) • The data is transferred from the buffer to the printer (at printer speed) • …allowing user to continue using the laptop // and allowing processor to continue processing • …instead of waiting for relatively slower printer • When the buffer is empty an interrupt is sent to the laptop • …requesting more data	**4**

9(d)	**1 mark** for naming a correct port **2 marks** for matching justification • USB • …has fast data transfer speeds for data (to the headset) • …is a universal/popular cable // universal standard • HDMI • …allows video and audio to be transferred (on the same cable) • …convenience of HDMI as no need for two cables	**3**

Question 10 · Oct/Nov 2024 Paper 12, original Q2 (a),(b),(c) [6 marks]

Question	Answer	Marks
10(a)	**1 mark** for each bullet point (**max 3**) e.g. • It is difficult to change / update the firmware by the user • Errors cannot be fixed easily // Troubleshooting / fault-finding / repairing is a specialist task / expensive • Functionality cannot be changed/extended easily // Cannot be easily adapted for another task • Faulty / outdated devices are often thrown away rather than repaired • … leading to e-waste	**3**

10(b)	**1 mark** from: • DRAM is less expensive to manufacture/purchase than SRAM in the embedded system • DRAM has a higher bit density per chip // more data can be stored per chip	**1**
10(c)	**1 mark** for each bullet point (**max 2**) • EPROM uses ultraviolet light to erase data whilst EEPROM uses an electrical signal to do this • EPROM has to be removed from the circuit board when changing the data whilst EEPROM remains in the circuit when the data is changed • EPROM erases all the data, EEPROM can erase parts of the data	**2**

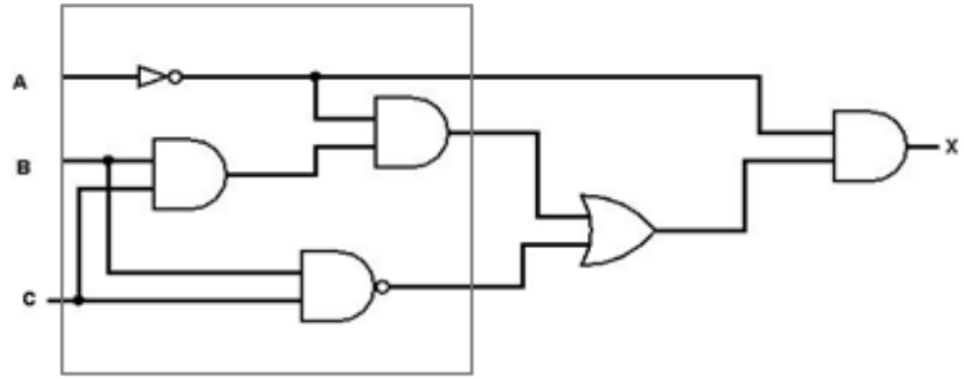
Question 11 · Oct/Nov 2024 Paper 12, original Q9 (a),(b) [4 marks]

Question	Answer	Marks
11(a)	**1 mark** for sensor **and** its use in this system • Infra-red sensor • Measure / check the height of the vehicle • Pressure sensor • Measure / check the weight of the vehicle	**2**

11(b)	**1 mark** for each bullet point (**max 2**) for a correct justification No marks for the identification of the system Monitoring system • Because there is no use of feedback // the warning sign is only an indicator • ... the output of the turning on of the sign does not affect the input of data from the sensors • The system does not have any actuators	**2**

Question 12 · Oct/Nov 2024 Paper 13, original Q1 (a),(b),(c) [5 marks]

Question	Answer	Marks
12(a)	**1 mark** for: Output is only 1 if one input is 1 and the other is 0 // Output is only 1 if both inputs are different // Output is only 0 if both inputs are the same	**1**
12(b)	**1 mark** each: • ((**A** NOR **B**) AND (**A** OR **B**)) • NAND (**B** AND **C**) **X** = ((**A** NOR **B**) AND (**A** OR **B**)) NAND (**B** AND **C**)	**2**

12(c)	**1 mark** for NOT gate, 2 correct AND gates and the NAND gate **1 mark** for OR gate and the final AND gate 	**2**

Question 13 · Oct/Nov 2024 Paper 13, original Q2 (a),(b) [6 marks]

Question	Answer	Marks
13(a)	**1 mark** for each difference e.g. • DRAM requires constant refreshing, but SRAM does not need to be constantly refreshed • DRAM generally has greater capacities per chip, but SRAM generally has lower capacities per chip • DRAM has slower access times than SRAM	**2**

13(b)	**1 mark** for each bullet point (**max 4**) • The computer and the hard disk drive transmit and receive at different speeds // The computer transfers data faster than the HDD can receive • The buffer is used for temporary storage • … so that the computer can transfer data to the buffer at the higher speed • ... and is not held up waiting for data to transfer • … and so that data is transferred to hard disk drive from the buffer at the slower rate	**4**